

# Expansions of the Icosian Triad

Where  $(\mathcal{I}, \mathcal{G}, \mathcal{C})$  appears in cosmology, biology, cascade dynamics,  
and the existence-closure programme

(Open research preprint — pre-peer-review draft)

Lee Smart

*Institute of Vibrational Field Dynamics*

contact@vibrationalfielddynamics.org

@vfd\_org

May 2026

## Abstract

The companion paper [1] (henceforth “Paper 1”) derived the icosian triad  $(\mathcal{I}, \mathcal{G}, \mathcal{C})$  from the geometry of the 600-cell vertex set  $V_{600}$ . The present paper catalogues the *expansions* of that triad — locations across the Vibrational Field Dynamics programme where the same structural objects already appear, often in seemingly unrelated contexts.

### Five expansion domains.

1. **Cascade ladder.**  $V_{600}$  is the  $H_4$  rung of the cascade [2, 3]. The  $\sigma$ -fixed eigenspace family (total dimension 94) and the  $\sigma$ -paired eigenspaces ( $E_1, E_2$  on one side,  $E_6, E_8$  on the other, with  $4 + 9 = 13$  dimensions per side) are numerically adjacent to the cascade’s finite-level refinement infrastructure [3, 4], but a vertex-level intertwiner between Paper 1’s  $\mathbb{C}[V_{600}]$  eigenspaces and the cascade’s auxiliary fibre  $F^0$  has not been constructed.
2. **Cosmology.** The hypersphere-cosmology release [5] uses the  $+6\varphi$  adjacency eigenspace of  $V_{600}$  (Paper 1’s  $E_1$ , dimension 4) as the substrate of the  $\Lambda$  dipole and the  $H_0$  correction. Paper 1 provides the formal definition of that eigenspace.
3. **Biology / consciousness.** Note G [6] states a consistency check between volatile-anaesthetic potency and a candidate microtubule-lattice realisation of the  $+6\varphi$  dipole eigenspace (Spearman  $\rho = +0.94$ ,  $n = 11$  volatile inhaled agents only). The substrate map remains conditional; the H-MT-Bridge cortical EEG prediction is the independence-promoting test. Paper 1 provides the underlying spectral object.
4. **Spectral bridge.** The Laplacian  $\lambda_L = 12$  eigenspace, which is Paper 1’s adjacency  $E_4$  (dimension 25,  $A_1$  eigenvalue 0), carries the Schläfli 5-coset spectral bridge of [7, 8]. Throughout this paper the convention is  $\lambda_L$  for Laplacian  $L = 12I - A_1$  and  $\lambda_{A_1}$  for adjacency.
5. **Existence programme.** The two involution conventions of [10] —  $\tau_{\text{ico}}$  with  $\dim \text{Fix} = 94$  and  $\tau_{\text{spec}}$  with  $\dim \text{Fix} = 116$  — relate to Paper 1 as follows.  $\tau_{\text{ico}}$  is identified with Paper 1’s Galois  $\sigma$ -action; its 94-dimensional fixed locus is the union of the  $\sigma$ -fixed  $A_1$ -eigenspaces.  $\tau_{\text{spec}}$  is the spectral involution acting as  $-1$  on the adjacency  $+6\varphi$  eigenspace (Paper 1’s  $E_1$ , dimension 4) and as identity on the remaining 116 dimensions; a geometric or vertex-level realisation of  $\tau_{\text{spec}}$  remains open.

**Status.** This paper makes no new mathematical claims about  $V_{600}$ . It is a positioning document. Cross-references are to existing public preprints and releases; predictions outside Paper 1 are stated as cited rather than as re-derived.

### Plain-language summary.

Paper 1 of this pair developed a single structural object — the *icosian triad* on  $V_{600}$ . Once that object is in hand, several parts of the broader programme that had been described with different languages turn out to be referring to the *same* mathematical content.

This paper points out those cross-identifications. It does not extend the mathematics; it shows where the mathematics already lives in published work, sometimes under different names.

The five recurrences listed in the abstract are: the cascade ladder, cosmology, biology, the 24–600 spectral bridge, and the existence programme’s involution conventions. Each is a separate paper or preprint in the public record. Each uses, implicitly or explicitly, the structure that Paper 1 derives in full.

### Expansion status table

The five expansions below relate to Paper 1 with different status. The following compact table records, for each: the cited source, the type of mathematical / empirical content, whether the cross-identification is a theorem, candidate substrate map, or numerical adjacency, and the principal open gap.

| Expansion           | Source   | Type                                                            | Status                                                                                                    | Principal open gap                                                               |
|---------------------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Cascade ladder      | [3, 4]   | Cited theorems (28-dim $W(H_4)$ -fixed $F^0$ , block Laplacian) | Spectral adjacency to Paper 1’s eigenspaces, no intertwiner                                               | Explicit intertwiner $F^0 \leftrightarrow \mathbb{C}[V_{600}]$                   |
| Cosmology           | [5]      | Empirical, 7 observables $< 0.5\%$ Planck 2018                  | Candidate substrate map; uses $E_1$ dipole numerically                                                    | Operator-level identification of cosmology $F$ with $\mathcal{C}_\varphi _{E_1}$ |
| Biology             | [6]      | Spearman $\rho = +0.94$ consistency check (volatile, $n = 11$ ) | Candidate microtubule substrate map; H-MT-Bridge conditional                                              | Cortical-EEG $R_{\text{phase}}$ pre-registered test                              |
| Spectral bridge     | [7–9]    | Cited theorems (exact $\mathbb{Q}(\sqrt{5})$ arithmetic)        | $\lambda_L = 12$ lift into Paper 1’s $E_4$                                                                | $A_5$ vs $W(H_4)$ equivariance distinction — settled in this paper               |
| Existence programme | [10, 11] | Cited definitions ( $\tau_{\text{ico}}, \tau_{\text{spec}}$ )   | $\tau_{\text{ico}} = \text{Paper 1’s } \sigma$ ; $\tau_{\text{spec}}$ spectrally defined as $-1$ on $E_1$ | Geometric / vertex-level realisation of $\tau_{\text{spec}}$                     |

This table is the load-bearing scope artefact for the cross-identifications below. Each subsequent section should be read against its row.

## 1 Introduction

The icosian triad  $(\mathcal{I}, \mathcal{G}, \mathcal{C})$  defined in Paper 1 is the structural object obtained by combining three simultaneous facts about the unit icosian group  $V_{600}$  inside the quaternion algebra over  $\mathbb{Q}(\sqrt{5})$ :

- a 9-class symmetric Bose–Mesner association scheme with  $A_1$ -spectrum in  $\mathbb{Z}[\varphi]$  and multiplicity sequence  $(1, 4, 9, 16, 25, 36, 9, 16, 4)$ ;
- a closure operator  $\mathcal{C}_\varphi = L_{V_{600}} + \varphi^{-2} \cdot I$  acting diagonally on the 9 eigenspaces;

- a quaternion norm map  $\mathcal{G} = N_H : \mathcal{I} \rightarrow \mathbb{Z}[\varphi]$  surjective onto totally positive  $\mathbb{Z}[\varphi]$ -primes, with  $\sigma$ -equivariance  $\sigma N_H = N_H \hat{\sigma}$ .

The total dimension  $\dim \text{Fix}(\sigma) = 94$  on the  $A_1$ -eigenspace decomposition is the load-bearing invariant.

The Vibrational Field Dynamics programme contains five public preprints (or release bundles) in which this triad already appears in disguise. The present paper catalogues these appearances. The mathematical content of each appearance is in the cited source; here we connect them via Paper 1.

## 2 Expansion 1: the cascade ladder

### Statement

The cascade ladder [2, 3] places  $V_{600}$  at the  $H_4$  rung, with downstream rungs at the 40-vertex root system (cf. [3]) and below. The cascade-selection-h4 paper [4] requires a  $W(H_4)$ -fixed subspace  $(F^0)^{W(H_4)}$  of dimension 28 on the fibre of each level of the finite-level refinement, and witnesses the (O1) closure obligation via the block Laplacian  $L_N^{\text{block}} = L_N^{\text{cl}} \otimes \text{id}_{F^0}$ .

### Connection

Paper 1's 9-class scheme contains the  $W(H_4)$ -representation structure of  $V_{600}$  at the  $\mathbb{C}[V_{600}]$  level. The 28-dimensional  $W(H_4)$ -fixed subspace of the cascade fibre  $F^0$  in [4] is a separate structural object from the  $\mathbb{C}[V_{600}]$  eigenspaces; an explicit intertwiner between the two spaces is not yet constructed.

What can be cleanly stated: cascade-selection-h4's block-Laplacian lowest non-zero eigenvalue at  $N = 0$ ,  $\lambda_+ = 9 - 3\sqrt{5} = 12 - 6\varphi$ , equals the Laplacian  $L = 12I - A_1$  eigenvalue corresponding to Paper 1's  $A_1$ -eigenvalue  $\lambda_1 = 6\varphi$  (eigenspace  $E_1$ , dimension 4). The  $\sigma$ -conjugate pair  $(E_1, E_8)$  of Paper 1's spectrum is the Galois partner that organises this lowest cascade band; explicit identification of the fibre  $F^0$  with subspaces of  $\mathbb{C}[V_{600}]$  is deferred to a follow-on note.

### Consequence

Paper 1 provides the full  $V_{600}$  spectral basis that organises the cascade's  $H_4$ -rung structure at the polytope level. The explicit intertwiner between cascade fibre  $F^0$  and  $\mathbb{C}[V_{600}]$  remains an open structural question.

## 3 Expansion 2: cosmology

### Statement

The hypersphere-cosmology release [5] derives the cosmological constant  $\Lambda$  as a  $\sigma$ -paired dipole correction to the  $E_8$ -substrate quadratic form, with the Planck-side numerics  $\Lambda$  within 0.078%,  $H_0$  within 0.45%, and  $\Omega_\Lambda$  within 0.083% of Planck 2018 inferred values, together with four further cosmological observables to  $< 0.5\%$  (seven total). The dipole construction uses the adjacency eigenvalue  $+6\varphi$  sector on  $V_{600}$  (Paper 1's  $E_1$ , dimension 4) and the complementary Laplacian value  $12 - 6\varphi$ .

## Connection (candidate substrate map)

The hypersphere-cosmology dipole sector is, numerically, the adjacency- $+6\varphi$  eigenspace of  $V_{600}$ , equivalently Paper 1's  $E_1$  (dimension 4). The Galois partner  $E_8$  (adjacency  $6 - 6\varphi$ , dimension 4) does not enter the cosmology paper's  $\Lambda$  formula directly; it is recorded by the Galois pairing as a separate object. The full identification of the cosmology source's operator  $F = \alpha R + \beta E - \gamma Q$  with a restriction of Paper 1's  $\mathcal{C}_\varphi$  is conditional on the cascade-substrate map and is not established at the operator level in either paper.

## Consequence

Paper 1 supplies the spectral basis ( $E_1$  at adjacency  $+6\varphi$ ) in which the cosmology source's dipole construction lives. The cosmology numerical success (94/94 verification tests) is consistent with the icosian triad as candidate substrate; a full operator-level identification would convert the consistency into a structural prediction and is open.

## 4 Expansion 3: biology / consciousness

### Statement

Note G [6] reports that the  $\sigma$ -paired closure-dipole class on  $V_{600}$ , when interpreted as a candidate microtubule-lattice dipole substrate, gives a consistency check between literature-DFT tubulin dipoles and clinical anaesthetic potencies: Spearman  $\rho = +0.94$  on  $n = 11$  volatile inhaled anaesthetics (the all-agent correlation including IV agents is much weaker,  $\rho \approx +0.5$ ). Note G frames this as a consistency check, not independent causal evidence. Note G also states the H-MT-Bridge conditional proposition: the cortical EEG  $R_{\text{phase}}$  ratio (sevoflurane / propofol) is at least 1.5 under equipotent dosing.

### Connection (candidate substrate map)

The dipole class of Note G is, in Paper 1's notation, the same adjacency- $+6\varphi$  eigenspace  $E_1$  (dimension 4) that appears in the cosmology source. The substrate map from Paper 1's  $E_1$  to the tubulin  $\alpha/\beta$ -dimer dipole array is a conditional candidate mapping; Note G explicitly does not validate it.

The Note G T15 result is a uniform  $V_{600}$ -leverage observation, not a residue-intersection result. The biological closure-control-node candidate residues identified in [6] arise from downstream symmetry breaking on the candidate substrate, not directly from the  $E_1$  basis.

### Consequence

Paper 1 supplies the spectral object ( $E_1$  at adjacency  $+6\varphi$ ) that Note G uses as its candidate substrate. The  $\rho = +0.94$  correlation is a consistency check on the candidate map (volatile-agent subset). The H-MT-Bridge cortical-EEG prediction is the independence-promoting test; until that test produces data, the substrate identification stays conditional.

## 5 Expansion 4: the 24-600 spectral bridge

### Statement

The three-paper 24-600 spectral bridge [7-9] establishes:

- the 5-coset decomposition  $V_{600} = 2T \sqcup 2T \cdot g \sqcup 2T \cdot g^2 \sqcup 2T \cdot g^3 \sqcup 2T \cdot g^4$  (right cosets;  $2T$  is non-normal in  $V_{600}$ ) into five 24-cell cosets;
- the selective lift of each local 24-cell Laplacian  $\lambda = 12$  eigenspace (dimension 2) into the global  $V_{600}$  Laplacian  $\lambda = 12$  eigenspace, of dimension 25, with exact  $\mathbb{Q}$ -rational characters;
- an explicit  $A_5$  action on the cosets with character  $5 \cdot Y_5 = 2 \cdot Y_5 \oplus 3 \cdot Y_5$ .

Throughout this section the symbol  $\lambda$  refers to the Laplacian  $L = 12I - A_1$ , not to the adjacency  $A_1$  used in Paper 1's spectrum table.

### Connection

Paper 1 supplies the full  $A_1$ -eigenstructure of  $V_{600}$ . Translating between conventions: the bridge's Laplacian eigenvalue  $\lambda_L = 12$  corresponds to Paper 1's  $A_1$ -eigenvalue  $\lambda_{A_1} = 0$ , hence to eigenspace  $E_4$  (dimension 25). The bridge's global 25-dimensional  $\lambda_L = 12$  space is therefore exactly Paper 1's  $E_4$ . Separately, Paper 1's  $E_0$  (dimension 1, the all-ones eigenspace at  $A_1$ -eigenvalue 12, equivalently Laplacian eigenvalue 0) accounts for the global constant row inside this picture.

The spectral bridge's selective lift corresponds to the  $A_5$ -equivariant projection onto  $E_4$  from each local 24-cell Laplacian's two-dimensional  $\lambda_L = 12$  subspace.

### Consequence

The spectral bridge is the  $E_4$ -projection of a broader 9-eigenspace structure, with  $E_0$  accounting separately for the constant row. Paper 1 places this in the wider context. The other seven eigenspaces of  $V_{600}$  ( $E_1, E_2, E_3, E_5, E_6, E_7, E_8$ ) carry the  $\sigma$ -paired dipole class, the antipodal  $\mathbb{Z}_2$ -grading, and the  $\sigma$ -paired 9-dimensional content, none of which is visible from the spectral bridge alone.

## 6 Expansion 5: the existence programme involutions

### Statement

The existence-life-closure programme [10, 11] uses two involutions on  $V_{600}$ :

$$\tau_{\text{ico}} : \dim \text{Fix} = 94, \quad \tau_{\text{spec}} : \dim \text{Fix} = 116.$$

Both commute with the closure operator  $\mathcal{C}_\varphi$  and both appear in theorem statements (the former) and simulation demonstrations (the latter) of the existence programme.

### Connection

$\tau_{\text{ico}}$  is the Galois action  $\sigma$  of Paper 1. Its fixed locus, restricted to the  $A_1$ -eigenspace decomposition, is

$$\text{Fix}(\sigma) = E_0 \oplus E_3 \oplus E_4 \oplus E_5 \oplus E_7 = \dim 1 + 16 + 25 + 36 + 16 = 94,$$

exactly matching the existence programme's load-bearing dimension.

### Consequence

Paper 1 derives the 94-dimensional  $\sigma$ -fixed locus structurally. The existence programme's  $\tau_{\text{ico}}$  convention is the same object.

## Spectral status of $\tau_{\text{spec}}$

The second involution  $\tau_{\text{spec}}$  with  $\dim \text{Fix} = 116$  is spectrally identified in [10]: it acts as  $-1$  on the adjacency  $+6\varphi$  eigenspace (Paper 1's  $E_1$ , dimension 4) and as identity on the remaining 116 dimensions. What remains open is a geometric / vertex-level realisation of this involution; that is a clean combinatorial follow-on.

## 7 What this paper does not do

- This paper does not extend Paper 1's mathematics. Quantitative claims used here are either cited from a prior source (cosmology, biology, cascade, bridge papers) or are direct arithmetic translations of Paper 1's spectrum into those sources' conventions. Cross-domain identifications are positioning claims, conditional on the substrate maps stated in each section.
- It does not address the Riemann Hypothesis or its generalisations.
- It does not propose new empirical predictions. The predictions referenced (e.g. Note G's H-MT-Bridge) remain at the status established in their source documents.
- It does not aim to be the canonical synthesis of the VFD programme; it is a localised positioning paper for the icosian triad's appearances in cosmology, biology, cascade dynamics, the 24–600 spectral bridge, and the existence-closure programme.

## 8 Open follow-on directions

1. Geometric identification of  $\tau_{\text{spec}}$  (existence programme) as an involution on  $V_{600}$  with  $\dim \text{Fix} = 116$ .
2. Construction of explicit Brandt / Hecke operators at higher prime powers; decomposition of  $\mathbb{C}[V_{600}]$  into Hecke-eigen-pieces.
3. Refinement of Note G's H-MT-Bridge from a consistency check into an operator-level prediction tied to Paper 1's  $E_1$  eigenspace (adjacency  $+6\varphi$ ) and its Galois partner  $E_8$ .
4. Numerical study of zeros of  $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1) \cdot C_2(s)$  (where  $C_2(s)$  is the explicit local-2 factor of Paper 1, Theorem 18) on the critical lines  $\text{Re}(s) = 1/2, 3/2$  (subject to classical RH and GRH for  $\chi_5$ ).
5. Comparison of the icosian theta with other definite quaternion-order theta functions over real quadratic fields (Hurwitz over  $\mathbb{Q}$ , Lipschitz over  $\mathbb{Q}$ , octahedral order over  $\mathbb{Q}(\sqrt{2})$ , ...) to characterise what is unique about the  $\mathbb{Q}(\sqrt{5})$  / icosian case.

## References

- [1] L. Smart, *The icosian triad on  $V_{600}$ : a geometry-first derivation*, Companion to this paper, Institute of Vibrational Field Dynamics, 2026.
- [2] L. Smart, *Cascade derivation: from  $E_8$  to the polytope ladder*, Open research preprint, Institute of Vibrational Field Dynamics, 2026.

- [3] L. Smart, *cascade-refinement-papers (P3 + P4 + cascade-refinement-compatible + cascade-mechanism + cascade-selection-h4)*, Five-paper bundle, Institute of Vibrational Field Dynamics, 2026.
- [4] L. Smart, *cascade-selection-h4: a witnessed (O1) discharge on the  $H_4$  branch*, In [3].
- [5] L. Smart, *Hypersphere Cosmology v1.1.0*, Public release, [github.com/vfd-org/hypersphere-cosmology](https://github.com/vfd-org/hypersphere-cosmology), 2026.
- [6] L. Smart, *Note G: Tubulin dipoles and the  $\sigma$ -paired closure-dipole class*, Open research preprint, Institute of Vibrational Field Dynamics, 2026.
- [7] L. Smart, *The Schläfli decomposition of the 600-cell*, In [8].
- [8] L. Smart, *the-24-600-spectral-bridge (three-paper bundle)*, Public release, [github.com/vfd-org/the-24-600-spectral-bridge](https://github.com/vfd-org/the-24-600-spectral-bridge), 2026.
- [9] L. Smart, *From Schläfli decomposition to spectral bridge: a first closure-projection channel in  $V_{600}$* , In [8].
- [10] L. Smart, *Existence as Closure*, Open research preprint, Institute of Vibrational Field Dynamics, 2026.
- [11] L. Smart, *Life as Closure*, Open research preprint, Institute of Vibrational Field Dynamics, 2026.